

Unit 18

CARBOXYLIC ACIDS AND DERIVATIVES

39 items · 99 marks

Syllabus sections in this unit

18.1 Carboxylic acids

18.2 Esters

Unit 18: Carboxylic acids and derivatives

1. M/J 26/P21/Q4/c

(c) Compound **B** reacts with an excess of LiAlH_4 to produce organic compound **C**.

Complete the equation in Figure 4.3 to show the reaction of **B** with an excess of LiAlH_4 to produce **C**.

[H] represents one atom of hydrogen from the reducing agent, LiAlH_4 .

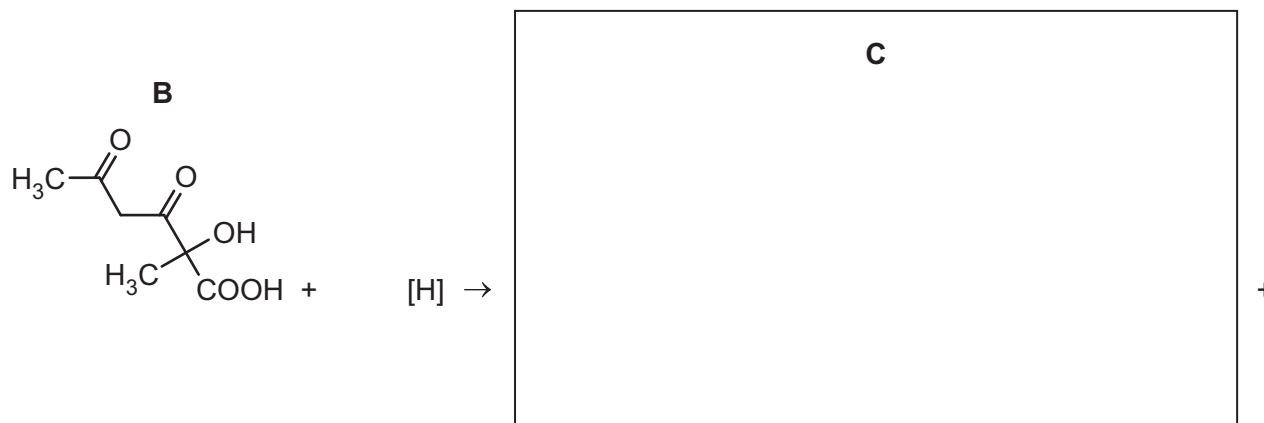


Figure 4.3

[2]

2. O/N 25/P22/Q4/d(i)

4 Fig. 4.1 shows how propane, C_3H_8 , can be converted to propanoic acid, $\text{CH}_3\text{CH}_2\text{COOH}$.

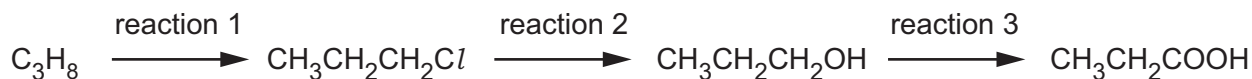


Fig. 4.1

(d) $\text{CH}_3\text{CH}_2\text{COOH}$ reacts with an unsaturated alcohol **R** to form unsaturated ester **S**.

(i) State the **type** of reaction that forms **S**.

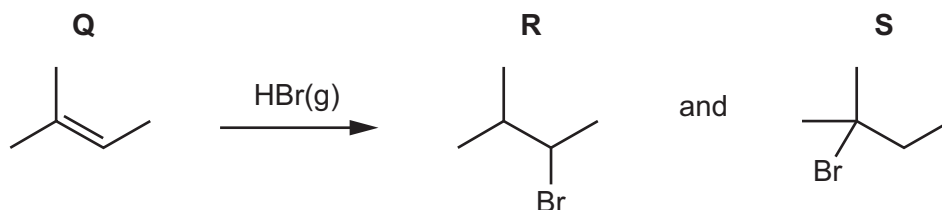
[1]

3. O/N 25/P24/Q4/b(iii)

4 Compounds **P** and **Q** are structural isomers.

**Fig. 4.1**

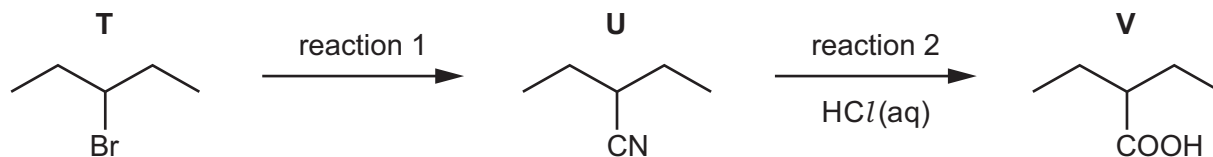
(v) **Q** reacts with HBr(g) to produce two structural isomers, **R** and **S**, as shown in Fig. 4.2.

**Fig. 4.2**

State and explain why isomer **S** is the major product of the reaction.

[2]

(b) Fig. 4.3 shows a synthesis starting from **T**, a different isomer of **R** and **S**.

**Fig. 4.3**

(iii) **V** reacts with propan-2-ol in the presence of a catalytic amount of H_2SO_4 to form organic compound **W**.

Complete Table 4.1 to give details of this reaction.

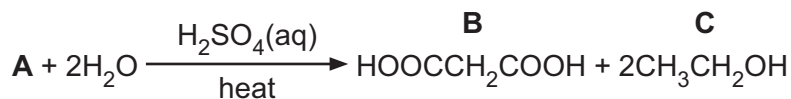
Table 4.1

	reaction of V with propan-2-ol
type of reaction	
functional group formed	
molecular formula of organic product W	

[3]

4. M/J 25/P21/Q5/a,b,c(i),e(i,ii)

- 5** Compound **A** contains the elements carbon, hydrogen and oxygen only.
When **A** is heated with $\text{H}_2\text{SO}_4(\text{aq})$, compounds **B** and **C** are produced, as shown in Fig. 5.1.

**Fig. 5.1**

- (a) Draw the structure of **A**. [2]
- (b) Name the type of reaction that occurs when **A** is heated with $\text{H}_2\text{SO}_4(\text{aq})$. [1]
- (c) When aqueous Na_2CO_3 is added to separate samples of **B** and **C**, effervescence is observed with **B** only.
- (i) Complete the equation to describe the reaction between **B** and an excess of aqueous Na_2CO_3 . [1]
- $$\text{HOOCCH}_2\text{COOH} + \text{Na}_2\text{CO}_3 \rightarrow$$
- (e) **B** reacts with an excess of reducing agent **R** to produce **D**.
An excess of PCl_5 is added to **D**.
- A vigorous reaction occurs.
 - Misty fumes are seen.
 - Organic compound **E** is produced.
- (i) Identify the formula of the reducing agent **R**. [1]
- (ii) Complete the equation to describe the reaction of **B** with an excess of **R**.
Use $[\text{H}]$ to represent one atom of hydrogen from **R**. [2]
- $$\text{HOOCCH}_2\text{COOH} + [\text{H}] \rightarrow$$

5. M/J 25/P22/Q4/a

- 4** (a) Propanoic acid, $\text{CH}_3\text{CH}_2\text{COOH}$, reacts with reducing agent **Q** to produce propan-1-ol, $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$.
- (i) Suggest the formula of reducing agent **Q**. [1]
- (ii) Complete the equation to show the reduction of propanoic acid to propan-1-ol.
Use $[\text{H}]$ to represent one atom of hydrogen from **Q**. [1]
- $$\text{CH}_3\text{CH}_2\text{COOH} + [\text{H}] \rightarrow$$

6. M/J 25/P23/Q5/b(ii,iii,iv)

5 (a) **W** is a colourless liquid.

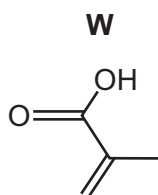


Fig. 5.1

(b) Fig. 5.2 shows two reactions of **W** to produce organic compounds **Y** and **Z**.

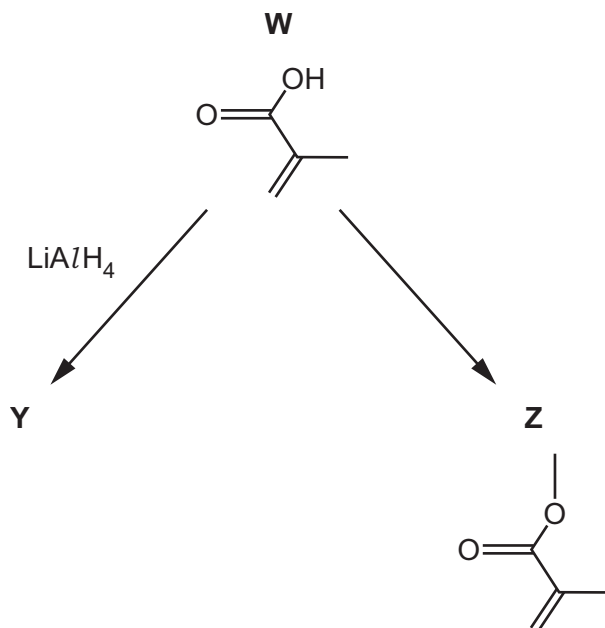


Fig. 5.2

(ii) **W** reacts with LiAlH_4 to produce **Y**.

Draw the structure of **Y**.

[1]

(iii) Identify the role of LiAlH_4 when it reacts with **W**.

[1]

(iv) Suggest the reagent and conditions required when **W** is converted into **Z**.

[1]

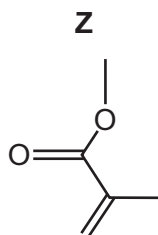


Fig. 5.3

7. M/J 25/P24/Q4/b,c,d(ii)

4 The skeletal formula of **E** is shown in Fig. 4.1.

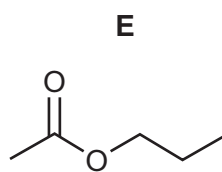


Fig. 4.1

(b) Name **E**.

[1]

(c) **E** is made when a carboxylic acid and an alcohol react together.

(i) Name the type of reaction that takes place during this esterification reaction.

[1]

(ii) Write an equation for the formation of **E** from a carboxylic acid and an alcohol. Use structural formulae to represent the organic species.

[1]

(iii) State the systematic name of the alcohol used to make **E**.

[1]

(d) **F** is an isomer of **E**.

F reacts with HCl(aq) to produce **G** and **H**.

G is a secondary alcohol.

H is also produced when HCl(aq) is added to CH_3CN .

(ii) Draw the displayed formula of **G**.

[2]

8. M/J 24/P23/Q5/d(iii,iv)

5 **(a)** **M** reacts to form **R** by the addition of one reagent, as shown in Fig. 5.1.

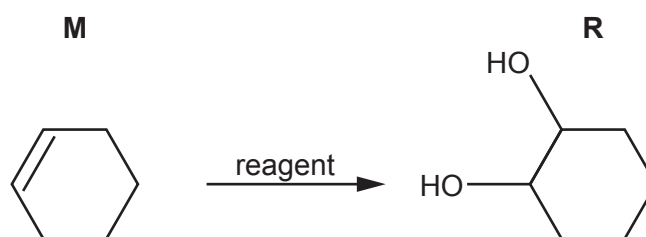


Fig. 5.1

Identify the reagent and conditions for this reaction.

[1]

(b) **R** is also made from **M** by two steps, as shown in Fig. 5.2.

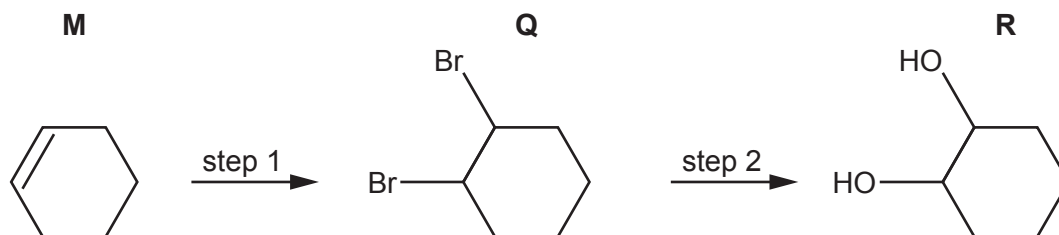


Fig. 5.2

(d) **Y** is made from **Q** in a three-step reaction.

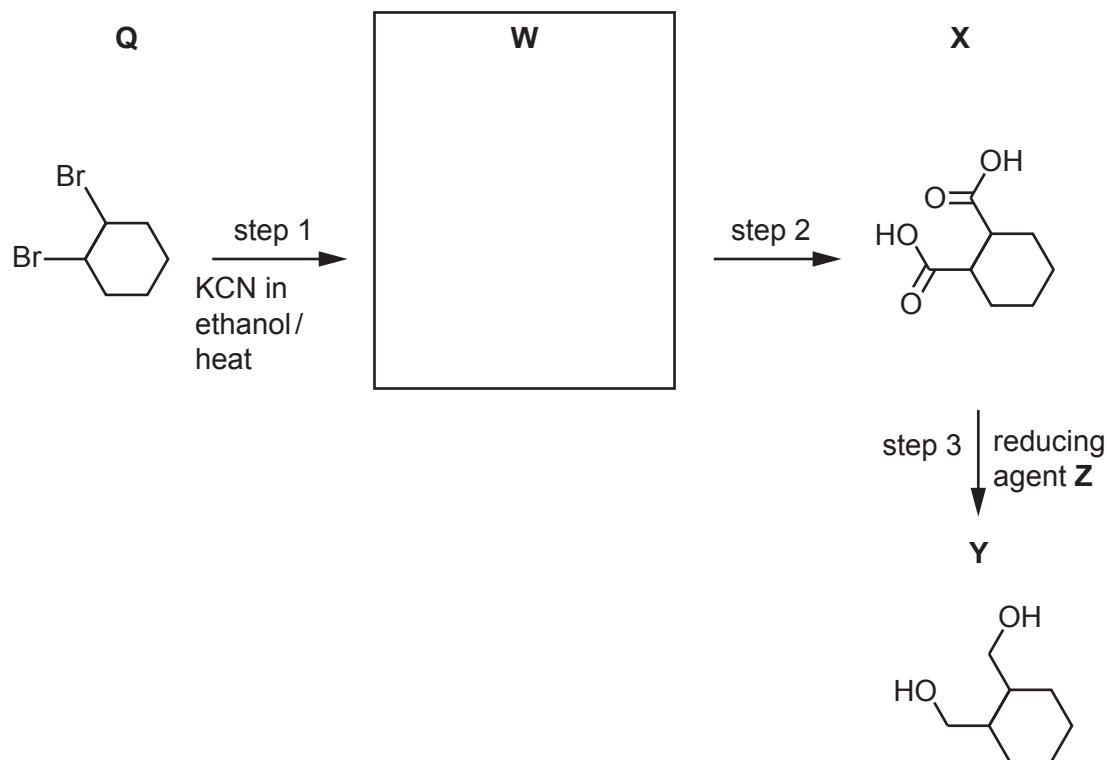


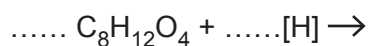
Fig. 5.4

(iii) In step 3, **X** reacts with reducing agent **Z** to produce **Y**.

Complete the equation for the reaction of **X** with **Z**.

Use a molecular formula to represent the organic product.

Use [H] to represent one atom of hydrogen from **Z**.



[1]

(iv) Identify **Z**.

[1]

9. O/N 23/P21/Q2/c

2 Barium hydroxide, $\text{Ba}(\text{OH})_2$, is a strong base used in inorganic and organic reactions.

Fig. 2.1 shows a reaction scheme involving $\text{Ba}(\text{OH})_2$.

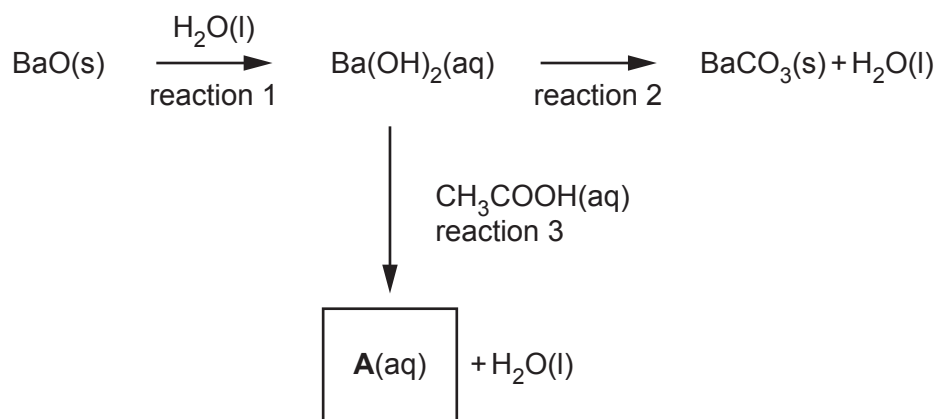


Fig. 2.1

(c) $\text{Ba}(\text{OH})_2$ is used to hydrolyse organic compounds.

Fig. 2.2 shows the reaction of **B** with $\text{Ba}(\text{OH})_2$, followed by acidification.

Draw the structures of the organic products of the process shown in Fig. 2.2.

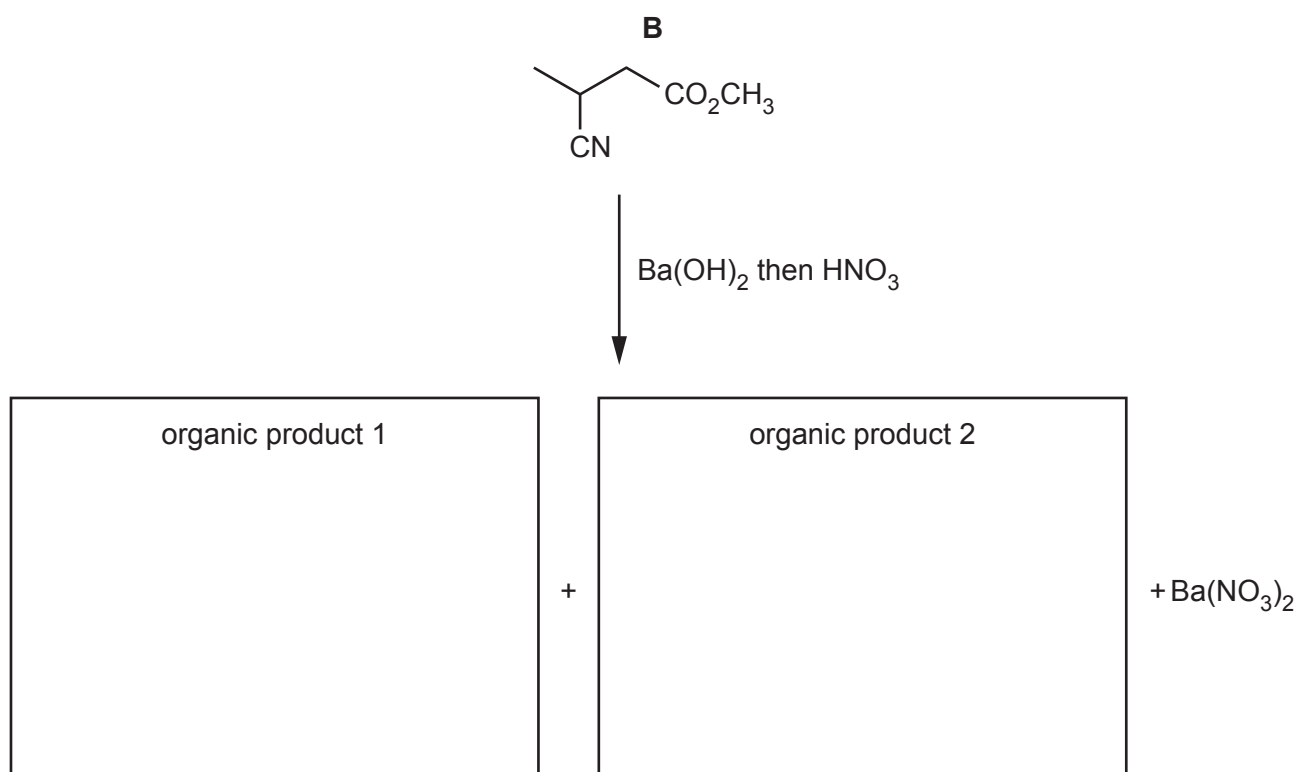


Fig. 2.2

10. O/N 23/P22/Q3/c(iii),d(i)

3 Phosphoric(V) acid, H_3PO_4 , is used in both inorganic and organic reactions.

(c) Fig. 3.1 shows a reaction scheme that involves H_3PO_4 in several reactions.

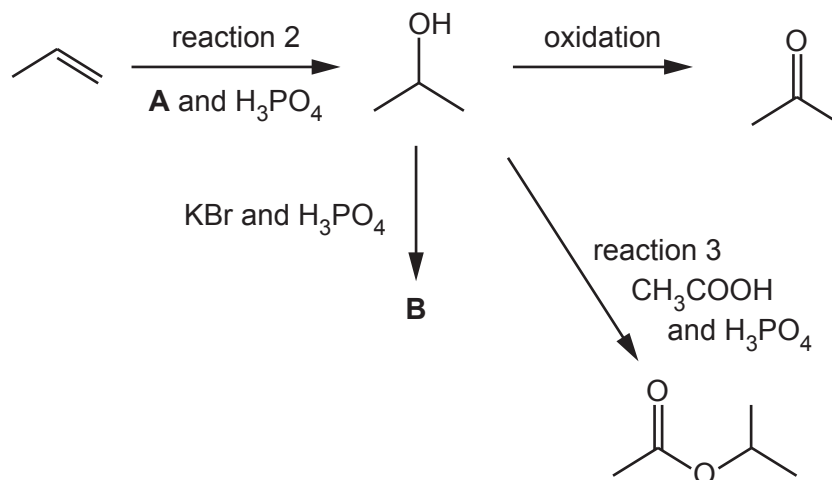


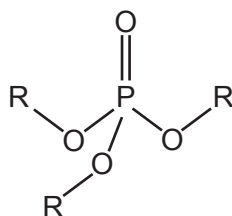
Fig. 3.1

(iii) Name the type of reaction that occurs in reaction 3.

[1]

(d) H_3PO_4 also reacts with alcohols to form organophosphates.

Organophosphates are compounds similar to esters. They have the general structure shown in Fig. 3.2.



R = alkyl group

Fig. 3.2

(i) Complete the equation to suggest the products of the reaction of H_3PO_4 with methanol, CH_3OH .

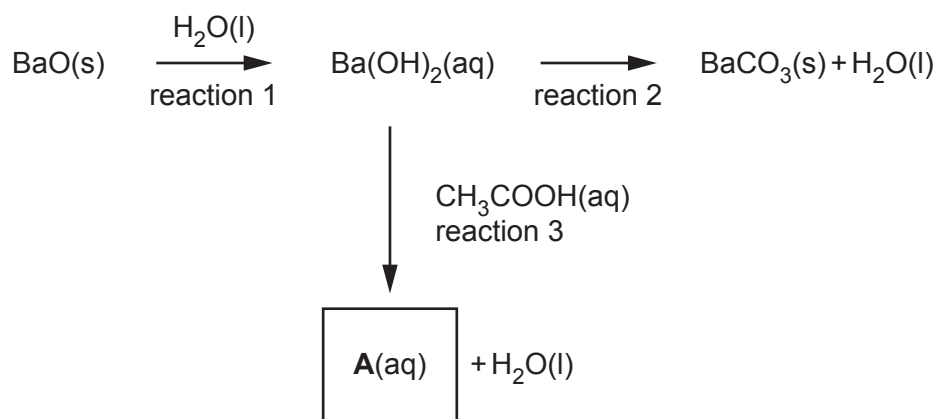


[1]

11. O/N 23/P23/Q2/c

2 Barium hydroxide, $\text{Ba}(\text{OH})_2$, is a strong base used in inorganic and organic reactions.

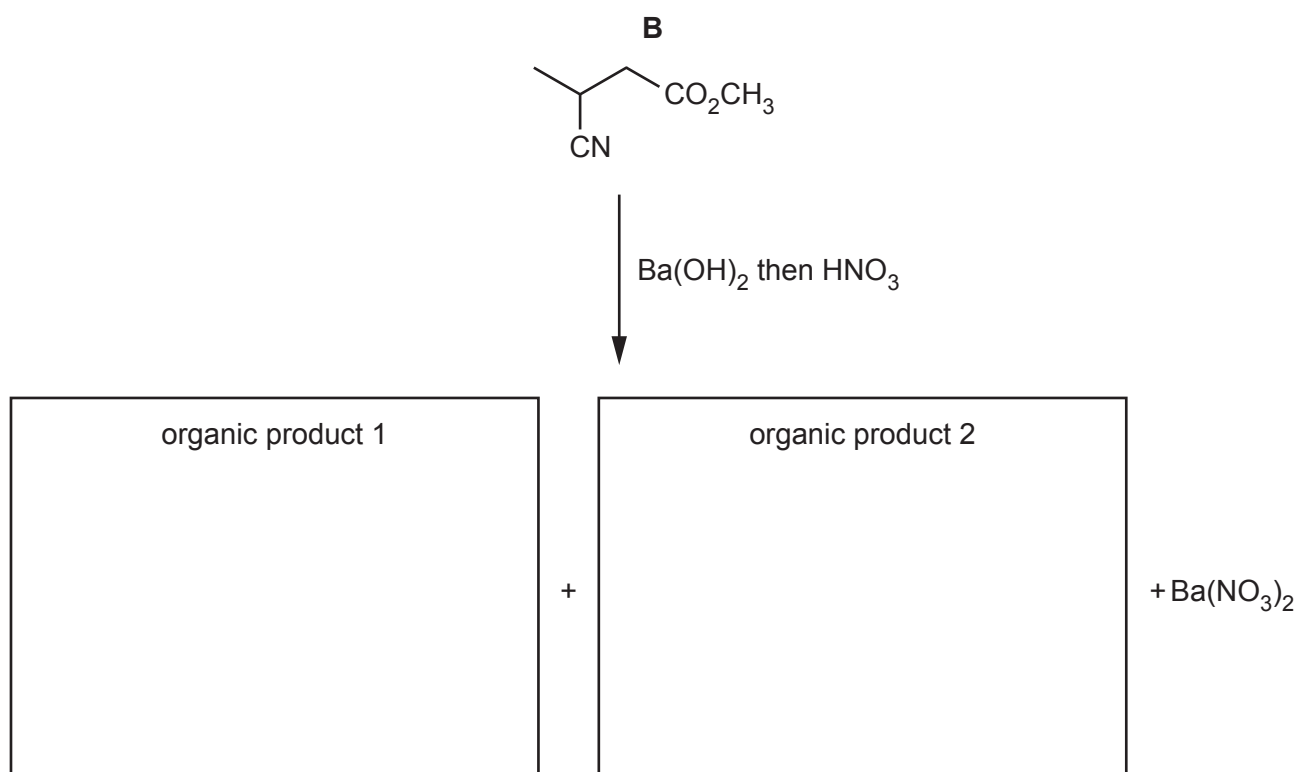
Fig. 2.1 shows a reaction scheme involving $\text{Ba}(\text{OH})_2$.

**Fig. 2.1**

(c) $\text{Ba}(\text{OH})_2$ is used to hydrolyse organic compounds.

Fig. 2.2 shows the reaction of **B** with $\text{Ba}(\text{OH})_2$, followed by acidification.

Draw the structures of the organic products of the process shown in Fig. 2.2.

**Fig. 2.2**

12. M/J 23/P21/Q5/c

(c) Biodiesel **T** is a fuel made from vegetable oil **R**. Fig. 5.1 shows the production of **T** from **R** in a two-step process.

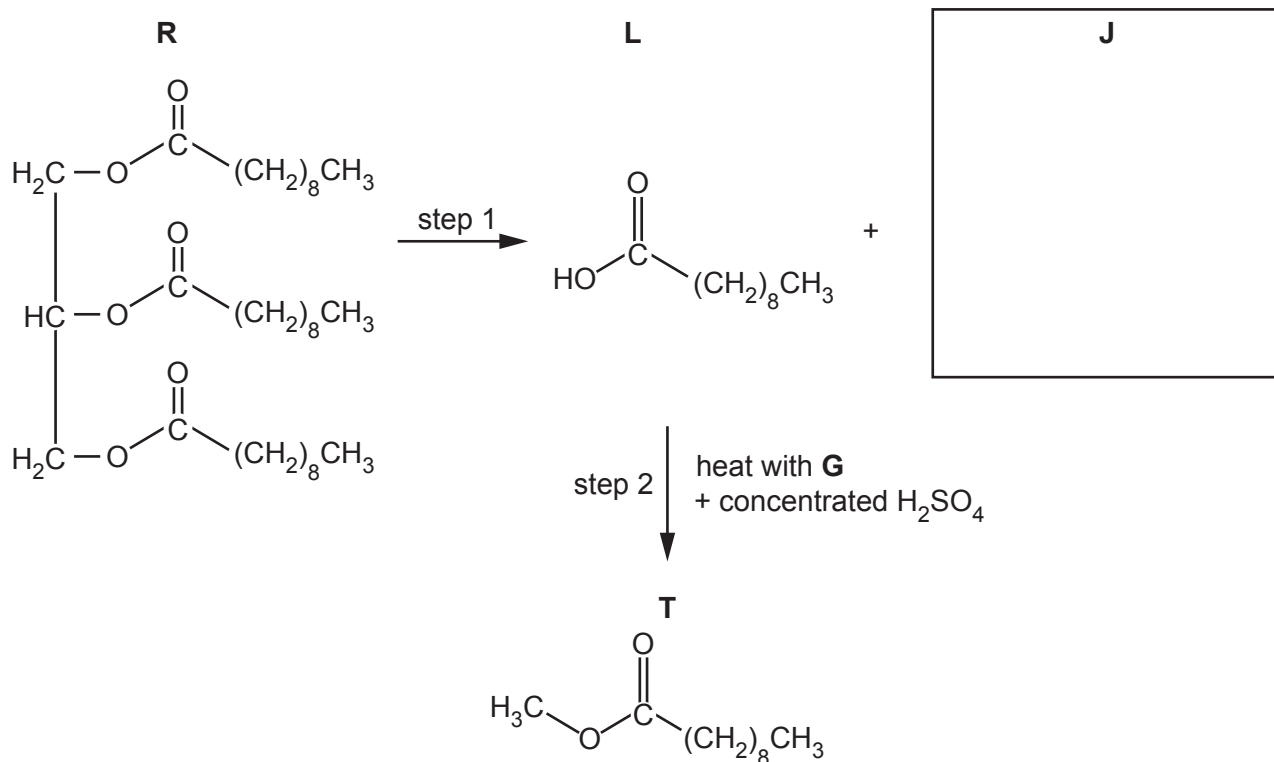
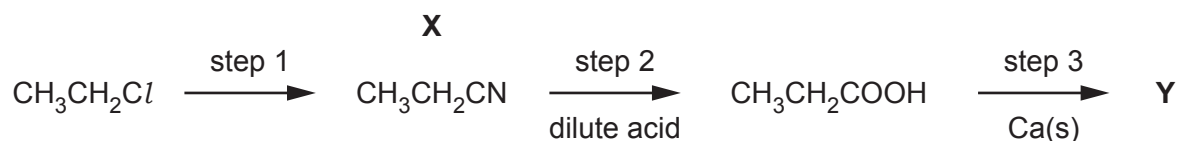


Fig. 5.1

- (i) In step 1 all three ester groups in **R** react. Suggest a suitable reagent and conditions for step 1. [1]
- (ii) Draw the structural formula of **J** in the box in Fig. 5.1. [1]
- (iii) Name the type of reaction that occurs in step 2. [1]
- (iv) Name organic reagent **G** used in step 2. [1]
- (v) **L** is called decanoic acid. Use systematic nomenclature to deduce the name of **T**. [1]

13. M/J 23/P22/Q3/a(iv)

- 3** Fig. 3.1 describes a sequence of reactions that can be used to produce a food additive, compound **Y**, from $\text{CH}_3\text{CH}_2\text{Cl}$.

**Fig. 3.1**

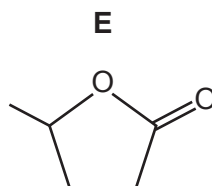
- (iv) In step 3, **Y** and a gas are produced.
Construct an equation for step 3.

[2]

14. M/J 23/P22/Q5/d(i,ii,iii)

- (d) **D** reacts in the presence of a sulfuric acid catalyst to form **E** and water.

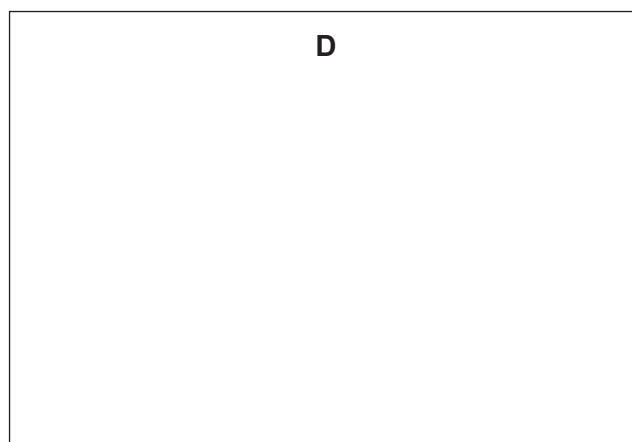
The structure of **E** is shown in Fig. 5.1.

**Fig. 5.1**

- (i) Name the functional group present in **E**.
- (ii) Identify the type of reaction that occurs when **D** reacts to form **E**.
- (iii) Draw the structure of **D** in the box.

[1]

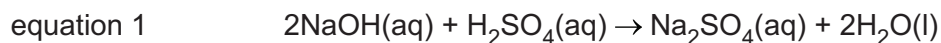
[1]



[1]

15. M/J 23/P23/Q3/c(i)

- 3** A neutralisation reaction occurs when NaOH(aq) is added to H₂SO₄(aq).



- (c)** The enthalpy change of neutralisation of CH₃COOH(aq) and NaOH(aq) is $-55.2 \text{ kJ mol}^{-1}$.

- (i)** Complete the equation for the reaction.

**16. O/N 22/P21/Q5/a(iv)**

- 5** Lactones are cyclic esters. Under suitable conditions, lactones form from molecules that have both an alcohol and a carboxylic acid functional group.

Equation 1 shows an example of the formation of a lactone.

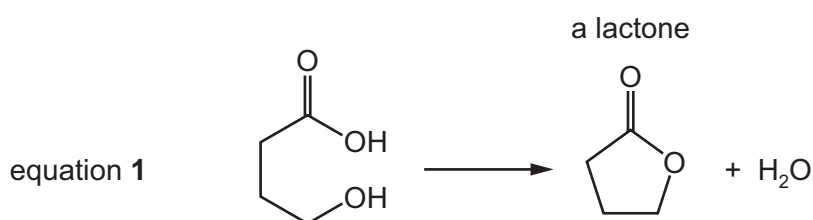


Fig. 5.1 shows the synthesis of lactone **P** from compound **M**.

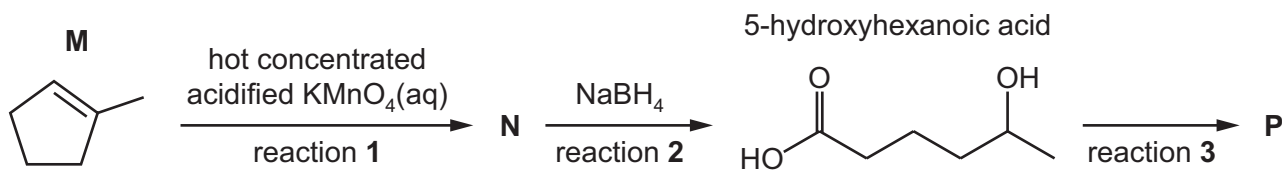


Fig. 5.1

- (iv)** Draw lactone **P**, the product of reaction 3.

[1]

17. O/N 22/P23/Q5/a(iv)

- 5** Lactones are cyclic esters. Under suitable conditions, lactones form from molecules that have both an alcohol and a carboxylic acid functional group.
Equation 1 shows an example of the formation of a lactone.

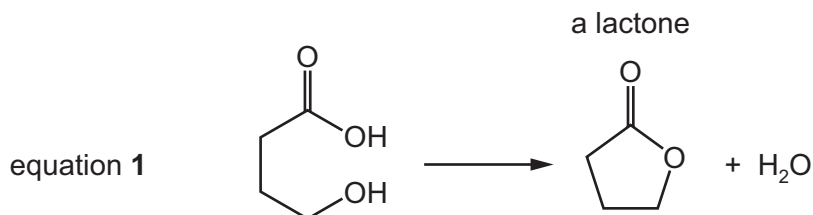


Fig. 5.1 shows the synthesis of lactone **P** from compound **M**.

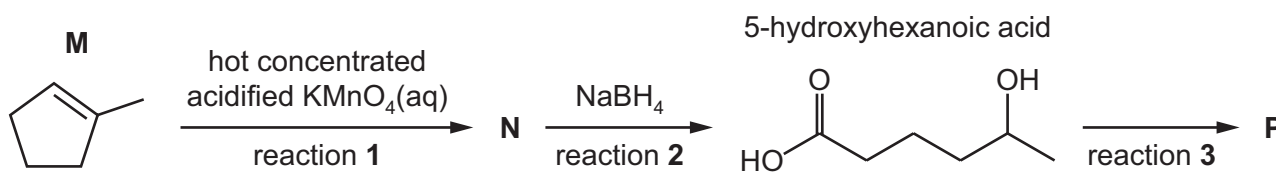


Fig. 5.1

- (iv)** Draw lactone **P**, the product of reaction 3.

[1]

18. M/J 22/P21/Q4/b(i)

- 4** Compound **V** is a liquid.

V contains 77.2% carbon, 11.4% hydrogen and 11.4% oxygen by mass.

V has a relative molecular mass of 280.

(b) **V** contains two types of functional group: a carboxylic acid and an alkene.

- (i)** Describe a chemical test and observation which confirms the presence of a carboxyl functional group.

[2]

19. M/J 22/P22/Q5/c(iii,iv)

5 Fig. 5.1 shows three reactions of 2-bromopropane, $\text{CH}_3\text{CH}(\text{Br})\text{CH}_3$.

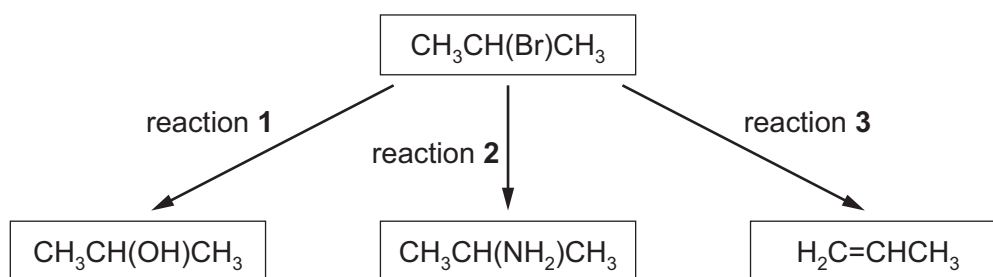


Fig. 5.1

(c) Fig. 5.2 shows how butan-1-ol can be made from 1-bromopropane in three steps.



Fig. 5.2

(iii) Step **3** is a reduction reaction.

Construct an equation for the reduction reaction in step **3**. Use $[\text{H}]$ to represent one atom of hydrogen from the reducing agent.

[1]

(iv) State the identity of a suitable reducing agent in step **3**.

[1]

20. M/J 22/P22/Q6/a,c(iii)

6 **Z** is a molecule which contains the elements carbon, hydrogen and oxygen only.

Z contains only alkene and carboxyl functional groups.

(a) Complete Table 6.1 by describing the observations that occur when two different reagents are added to separate samples of **Z(aq)**.

Table 6.1

reagent added to Z(aq)	observation
$\text{Br}_2(\text{aq})$	
$\text{Na}_2\text{CO}_3(\text{s})$	

[2]

(c) Fig. 6.1 shows the mass spectrum of **Z**.

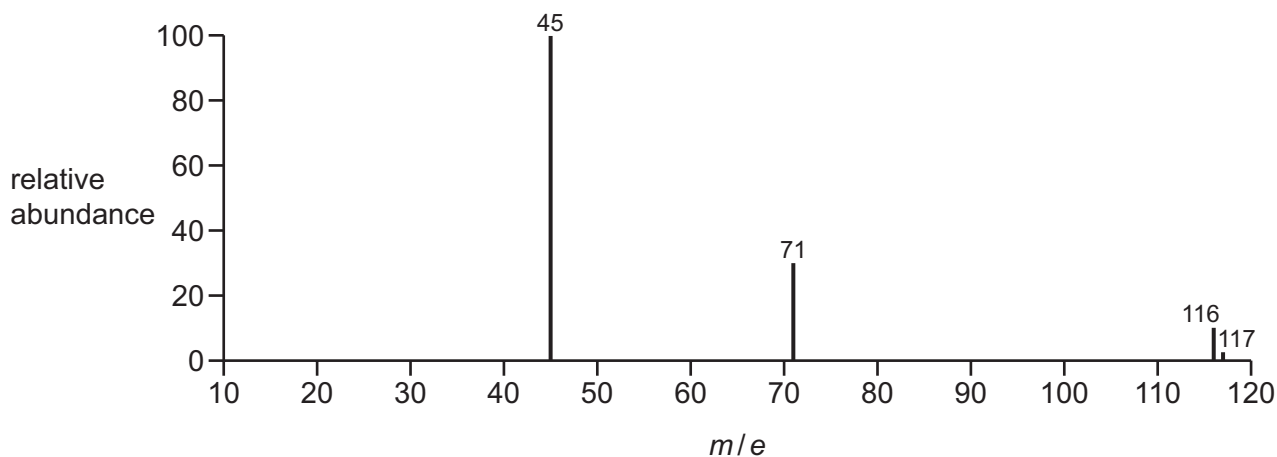


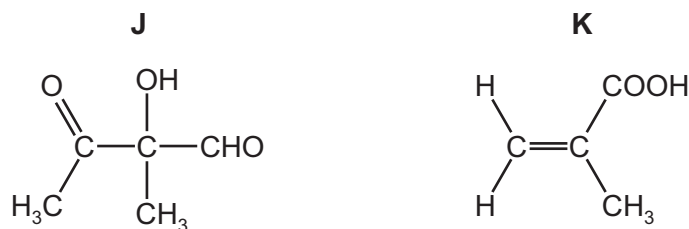
Fig. 6.1

(iii) Suggest the structure of **Z** using relevant information from Table 6.1, **(b)** and **(c)**.

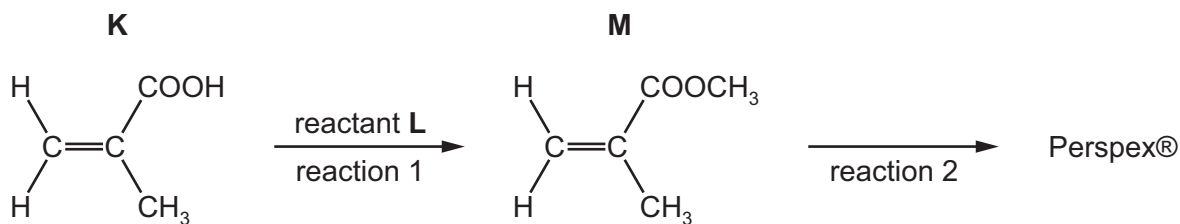
[1]

21. MAR 22/P22/Q4/b(i)

4 Compounds **J** and **K** are found in plant oils.

**Fig. 4.1**

(b) **K** is used to make the addition polymer Perspex®. A synthesis of Perspex® is shown in Fig. 4.2.

**Fig. 4.2**

(i) Identify **L**. State the conditions required for reaction 1.

L =

conditions =

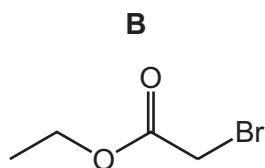
[2]

22. O/N 21/P21/Q4/c

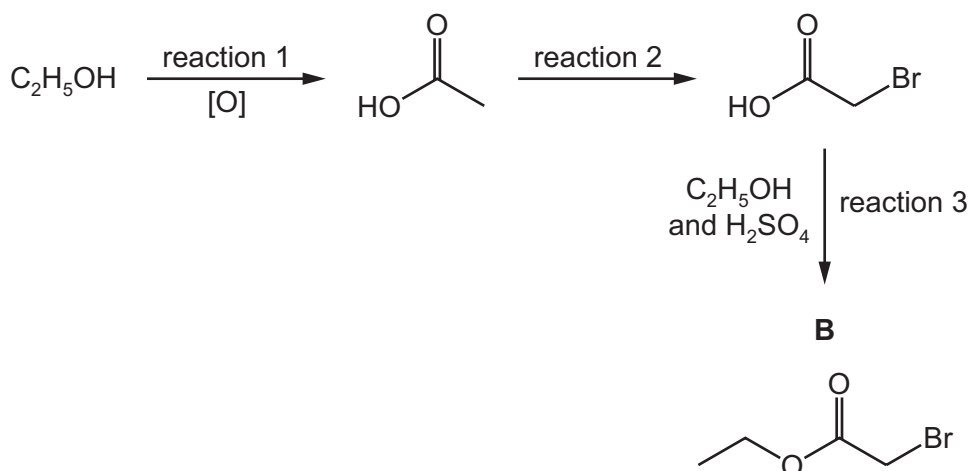
also Unit 16 (1 mark); this unit 2 marks

Data Booklet needed

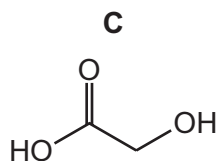
4 Compound **B** is a liquid with a fruity smell.



The reaction scheme shows how **B** can be made from ethanol, C₂H₅OH.



- (b) Reaction 2 needs to take place in the absence of water to prevent formation of compound **C**.



If **C** is present in the reaction mixture of reaction 3, a different compound, compound **D**, will also form. Compound **D** has two identical functional groups.

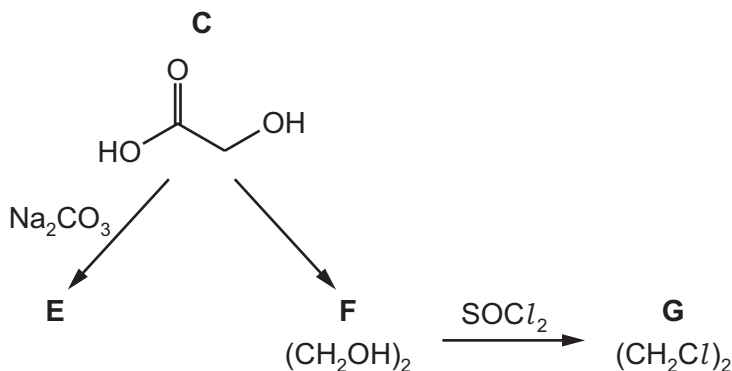
The infrared spectrum of **D** shows strong absorptions at 1100 cm^{-1} and 1720 cm^{-1} , but no absorption due to O–H bonds.

Use the *Data Booklet* to identify the functional group present in **D**.

Explain your answer as fully as you can.

[3]

- (c) Some other reactions of **C** are shown.



- (i) Draw the structure of **E**.

[1]

- (ii) Suggest why NaBH_4 is not a suitable reagent to make **F**, $(\text{CH}_2\text{OH})_2$, from **C**. Explain your answer.

[1]

- (iii) Construct an equation for the reaction of $(\text{CH}_2\text{OH})_2$ with SOCl_2 to form **G**, $(\text{CH}_2\text{Cl})_2$.

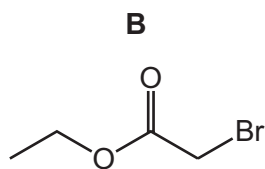
[1]

23. O/N 21/P23/Q4/c

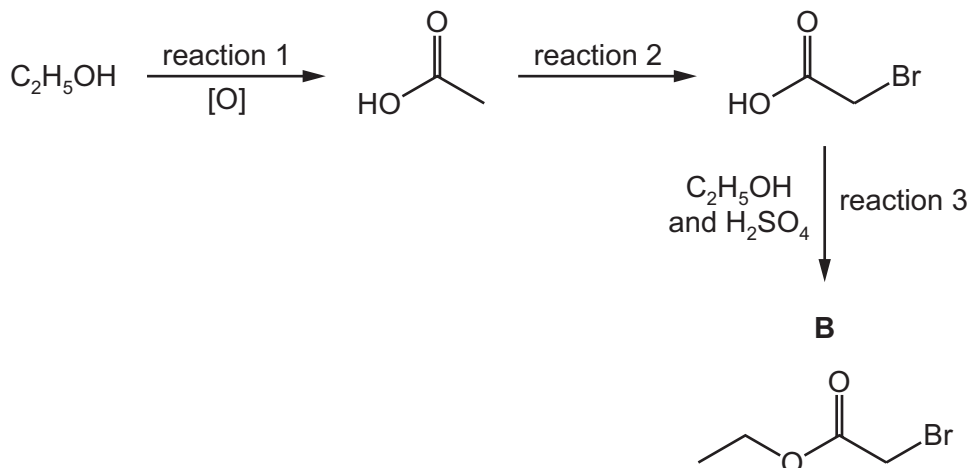
also Unit 16 (1 mark); this unit 2 marks

Data Booklet needed

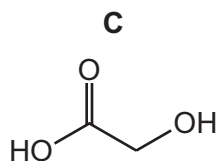
- 4** Compound **B** is a liquid with a fruity smell.



The reaction scheme shows how **B** can be made from ethanol, $\text{C}_2\text{H}_5\text{OH}$.



(b) Reaction 2 needs to take place in the absence of water to prevent formation of compound **C**.



If **C** is present in the reaction mixture of reaction 3, a different compound, compound **D**, will also form. Compound **D** has two identical functional groups.

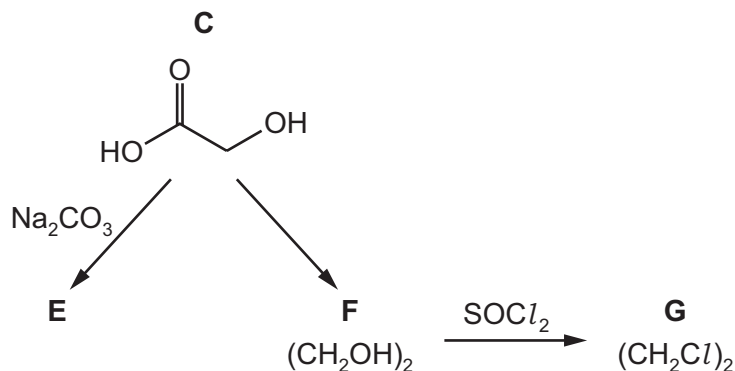
The infrared spectrum of **D** shows strong absorptions at 1100 cm^{-1} and 1720 cm^{-1} , but no absorption due to O–H bonds.

Use the *Data Booklet* to identify the functional group present in **D**.

Explain your answer as fully as you can.

[3]

(c) Some other reactions of **C** are shown.



(i) Draw the structure of **E**.

[1]

- (ii) Suggest why NaBH_4 is not a suitable reagent to make **F**, $(\text{CH}_2\text{OH})_2$, from **C**. Explain your answer.

[1]

- (iii) Construct an equation for the reaction of $(\text{CH}_2\text{OH})_2$ with SOCl_2 to form **G**, $(\text{CH}_2\text{Cl})_2$.

[1]

24. M/J 21/P21/Q5/c(i)

- (c) Propene is separated from the mixture and heated in air in the presence of a catalyst. Propene is oxidised to **X**, which contains two functional groups.

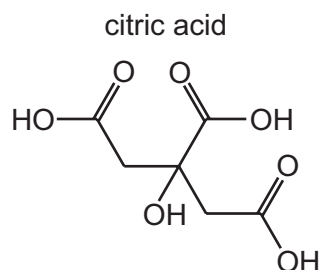
- (i) Effervescence is seen when $\text{Na}_2\text{CO}_3(\text{aq})$ is added to **X**.

Identify the functional group present in **X** which is responsible for this observation.

[1]

25. M/J 21/P23/Q5/a

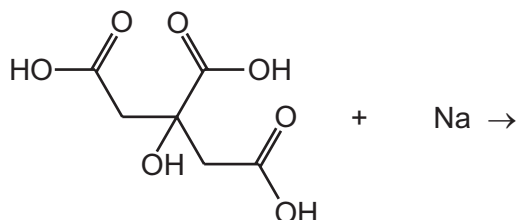
- 5** Compound **X** contains the same functional groups as citric acid.



The table describes some of the similarities and differences between citric acid and compound **X**.

	citric acid	X
chiral centre	no	yes
reaction with Na	fizzing	fizzing
reaction with $\text{H}^+/\text{Cr}_2\text{O}_7^{2-}$	remains orange	orange to green

- (a) Complete the equation to show the reaction of excess sodium with citric acid. Show the skeletal structure of the product.



[3]

26. M/J 20/P21/Q5/a(iii)

5 (a) Below is a list of species which can react with organic compounds.



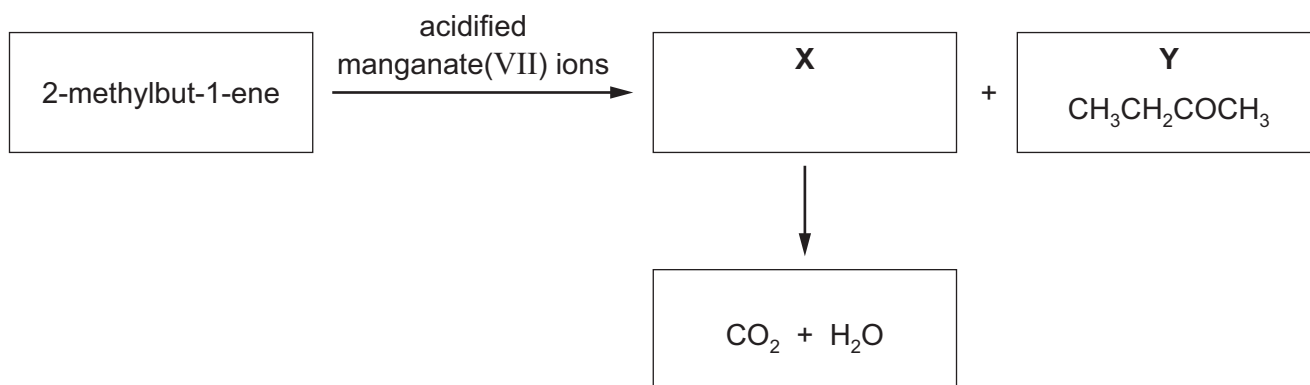
(iii) From the list, identify a species which can be used to distinguish between solutions of propanoic acid and propan-1-ol. Describe any relevant observations.

[2]

27. M/J 20/P21/Q6/f,g(ii)

6 2-methylbut-1-ene reacts with acidified manganate(VII) ions, under specific conditions, to produce two organic compounds **X** and **Y**.

X immediately reacts with the acidified manganate(VII) ions to form carbon dioxide and water. **Y** has the structural formula $\text{CH}_3\text{CH}_2\text{COCH}_3$.



(f) Propanoic acid, $\text{CH}_3\text{CH}_2\text{CO}_2\text{H}$, is reduced by LiAlH_4 .

(i) Write an equation to show this reaction. Use $[\text{H}]$ to represent an atom of hydrogen from the reducing agent.

[1]

(ii) Name the organic product formed in this reaction.

[1]

(g) Organic compound **W** is an ester which is a structural isomer of propanoic acid.

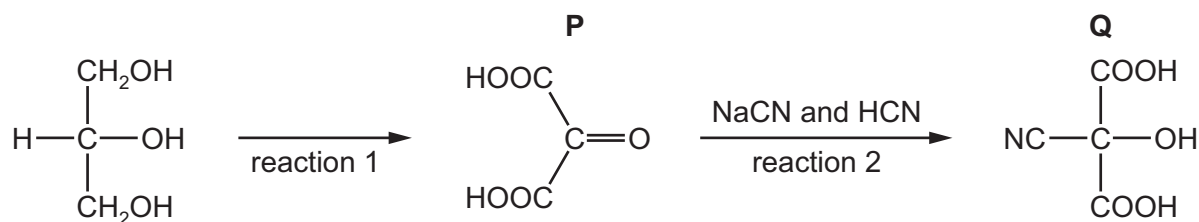
(ii) Draw a possible structure of **W**.

[1]

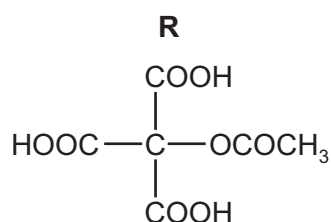
28. MAR 20/P22/Q3/a(v)

3 Glycerol, $\text{CH}_2(\text{OH})\text{CH}(\text{OH})\text{CH}_2\text{OH}$, is widely used in the food industry and in pharmaceuticals.

(a) A series of reactions starting from glycerol is shown.



(v) When **Q** is heated with excess aqueous ethanoic acid in the presence of a catalytic amount of sulfuric acid, two reactions take place to form compound **R**.



Identify the two types of reaction that occur.

1

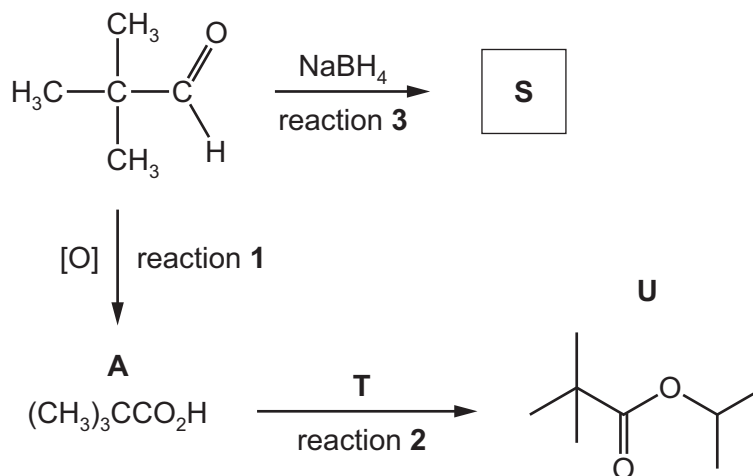
2

[2]

29. O/N 19/P21/Q4/b(iv)

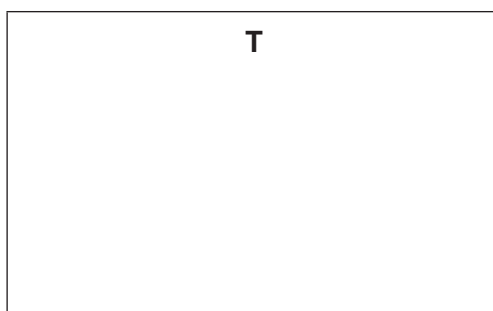
4 $(\text{CH}_3)_3\text{CCHO}$ is used in the synthesis of some antibiotics.

(b) Two reaction sequences are shown.



(iv) Draw the structure of the organic molecule **T** that reacts with **A**, $(\text{CH}_3)_3\text{CCO}_2\text{H}$, in reaction **2**, to form **U**.

Suggest a catalyst for reaction **2**.

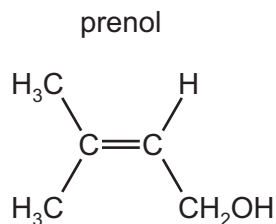


catalyst

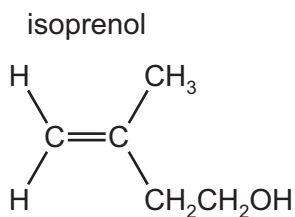
[2]

30. O/N 19/P22/Q4/d(iii)

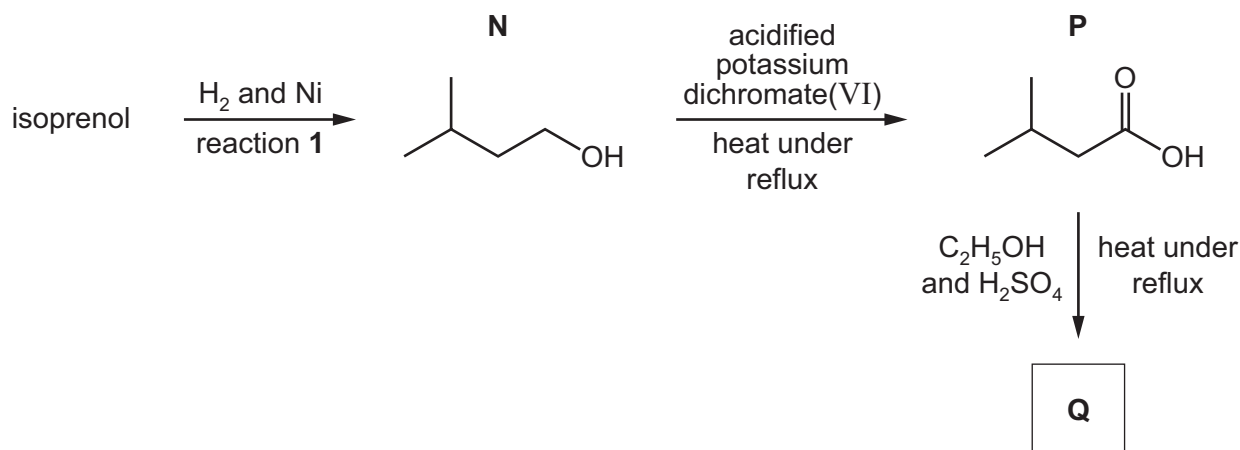
- 4 Prenol is a naturally occurring organic molecule found in many fruits. It contains both an alkene and an alcohol functional group.



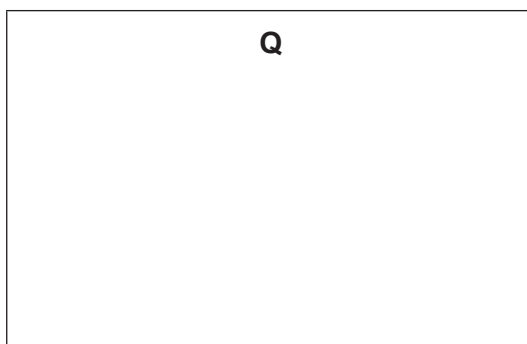
- (d) Isoprenol is a structural isomer of prenol.



The series of reactions shows how isoprenol can be used to form **Q**, a sweet-smelling liquid.



- (iii) Draw the **skeletal** formula of **Q** and suggest one commercial use of **Q**.

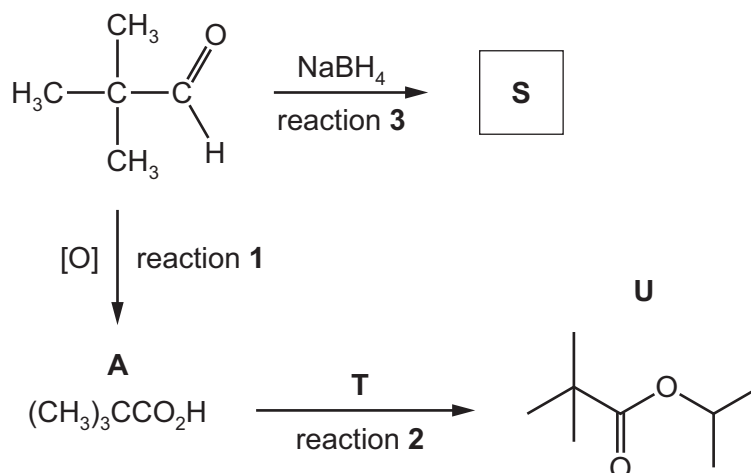


commercial use

31. O/N 19/P23/Q4/b(iv)

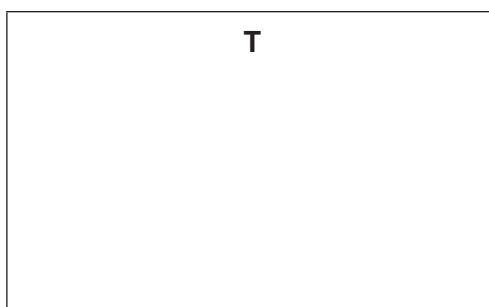
4 $(\text{CH}_3)_3\text{CCHO}$ is used in the synthesis of some antibiotics.

(b) Two reaction sequences are shown.



(iv) Draw the structure of the organic molecule **T** that reacts with **A**, $(\text{CH}_3)_3\text{CCO}_2\text{H}$, in reaction 2, to form **U**.

Suggest a catalyst for reaction 2.



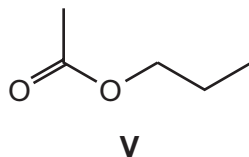
catalyst

[2]

32. M/J 19/P22/Q5/a(i,ii,iii),d

5 Many naturally occurring esters are used as flavourings in food.

(a) The structure of ester **V** is shown.



(i) Name **V**.

[1]

V reacts with a reagent to form a salt of a carboxylic acid and an alcohol.

(ii) Identify a reagent that could be used in this reaction.

[1]

(iii) Draw the displayed formula of the alcohol made during this reaction.

[1]

(d) Sodium carbonate solution reacts with methanoic acid.

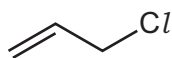
Write the equation for this reaction.

[1]

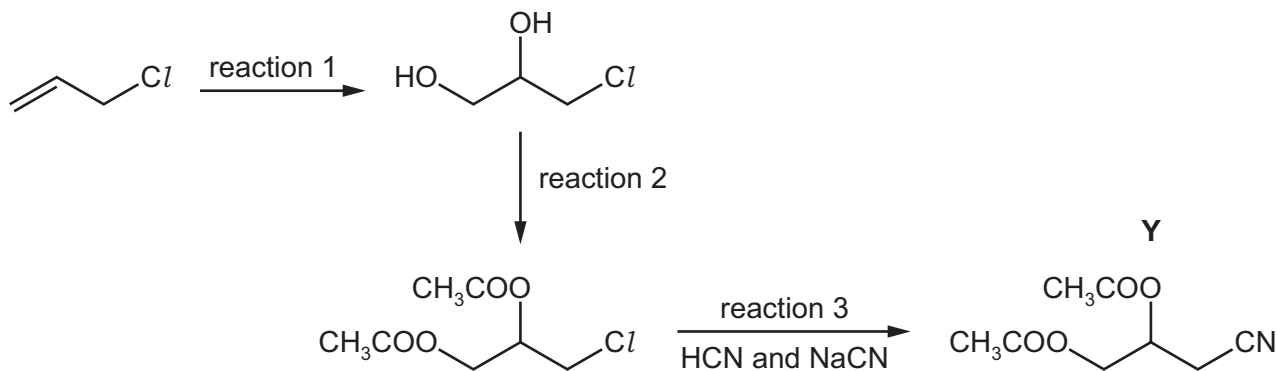
33. MAR 19/P22/Q4/c(ii,iv)

- 4 Allyl chloride is an important chemical used in the manufacture of plastics, pharmaceuticals and pesticides.

allyl chloride



(c) A series of reactions starting from allyl chloride is shown.



- (ii) In reaction 2, the organic product of reaction 1 is mixed with concentrated H_2SO_4 and an organic acid, and then heated under reflux.

State the role of the concentrated H_2SO_4 . Identify the organic acid used.

role of the concentrated H_2SO_4

identity of the organic acid

[2]

- (iv) The organic product of reaction 3 is **Y**.

Y can be hydrolysed using excess aqueous H_2SO_4 to form **Z**.

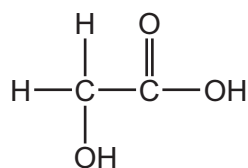
The molecular formula of **Z** is $\text{C}_4\text{H}_8\text{O}_4$.

Draw the structure of **Z**.

[2]

34. O/N 18/P21/Q4/d(ii)

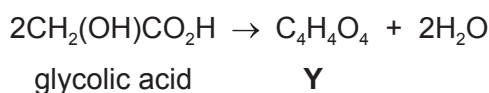
4 The structure of glycolic acid is shown.



glycolic acid

(d) When glycolic acid is heated in the presence of a sulfuric acid catalyst, a new compound, **Y**, $\text{C}_4\text{H}_4\text{O}_4$, is formed.

The equation for the reaction is given.

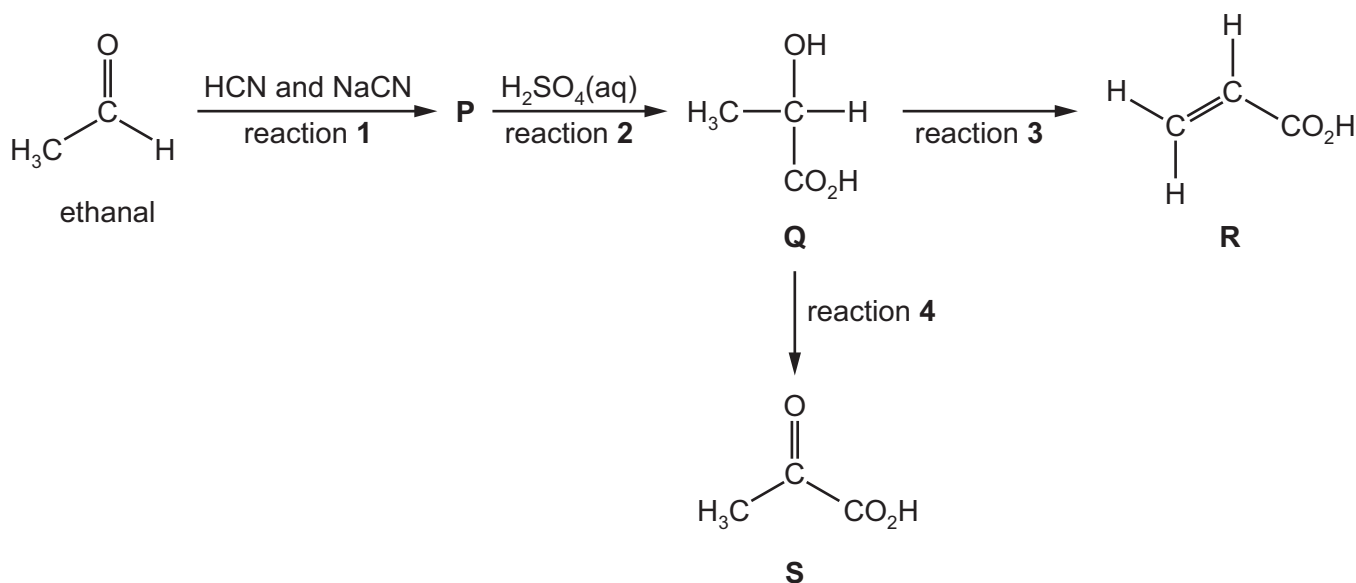


(ii) Suggest a structure for **Y**.

[2]

35. O/N 18/P22/Q4/e(i)

4 The diagram shows a reaction sequence starting from ethanal.



(e) (i) Compound **S**, $\text{CH}_3\text{COCO}_2\text{H}$, can be reduced by LiAlH_4 .

Complete the equation using structural formulae to represent this reaction.
Use $[\text{H}]$ to represent the reducing agent.

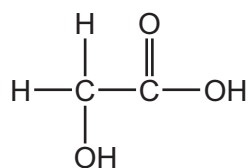


[2]

Other reducing agents containing Group 1 metal cations include LiBH_4 , NaBH_4 and KBH_4 .
The strength of the reducing agent depends on the size of its cation.

36. O/N 18/P23/Q4/d(ii)

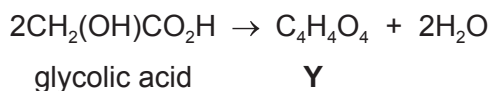
4 The structure of glycolic acid is shown.



glycolic acid

(d) When glycolic acid is heated in the presence of a sulfuric acid catalyst, a new compound, **Y**, $\text{C}_4\text{H}_4\text{O}_4$, is formed.

The equation for the reaction is given.



(ii) Suggest a structure for **Y**.

[2]

37. M/J 17/P22/Q4/b(ii)

(b) The alcohols **C** and **D** are isomers of each other with molecular formula $\text{C}_4\text{H}_{10}\text{O}$. Both isomers are branched.

When **C** is heated under reflux with acidified potassium dichromate(VI) no colour change is observed.

When **D** is heated under reflux with acidified potassium dichromate(VI) the colour of the mixture changes from orange to green and **E**, $\text{C}_4\text{H}_8\text{O}_2$, is produced.

E reacts with aqueous sodium carbonate to form carbon dioxide gas.

(ii) Write the equation for the reaction between **E** and aqueous sodium carbonate.

[1]

38. MAR 17/P22/Q3/d(i)

(d) $\text{CH}_3(\text{CH}_2)_3\text{CO}_2\text{H}$ is a colourless liquid with an unpleasant odour.

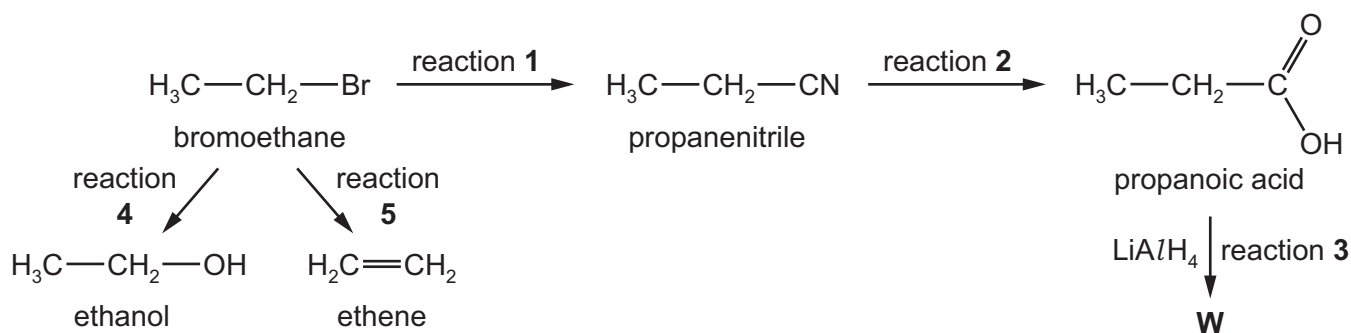
It reacts with methanol in the presence of an acid catalyst to produce an organic product **V**, which has a pleasant fruity smell.

(i) Name **V**.

[1]

39. M/J 16/P21/Q5/b(i),d

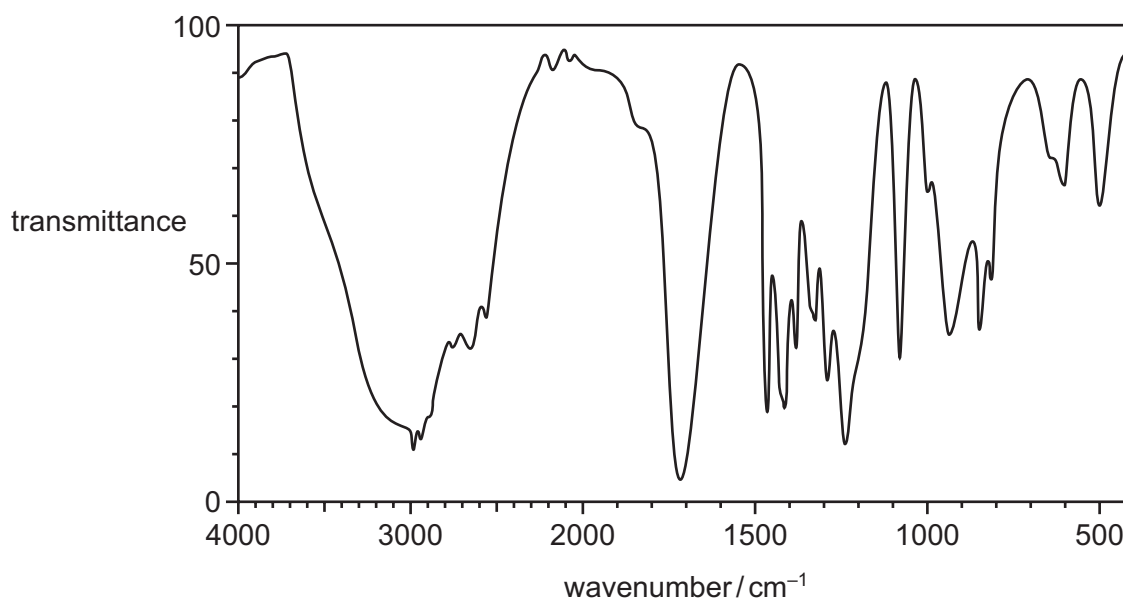
5 A reaction sequence is shown.



(b) (i) Give the name of the type of reaction involved in reaction 3.

[1]

The infra-red spectrum of the propanoic acid produced by reaction 2 is shown.



(d) Under appropriate conditions, ethanol and propanoic acid undergo a condensation reaction.

(i) State the condition necessary for the reaction.

[1]

(ii) Draw the skeletal formula of the organic product of this reaction.

[1]

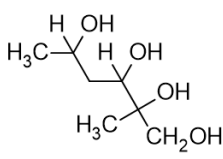
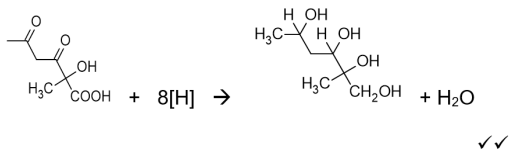
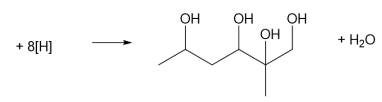
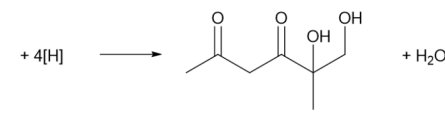
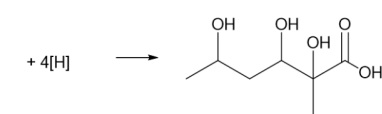
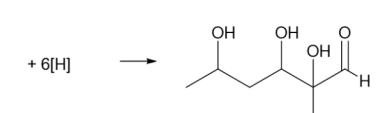
(iii) Name the organic product of this reaction.

[1]

Question 5 continues over the page.

Answers Section

1. M/J 26/P21/Q4/c

4(c)	M1  M2 equation fully correct  M2 equation fully correct  ✓✓	2 M1 A any unambiguous structure of C M2 stoichiometry based on correct structure C A ecf on M2 if incorrect structure is correctly balanced – as long as structure contains 7 carbon atoms common ECFs for 1 mark  M2 equation fully correct  M2 equation fully correct  ✓✓
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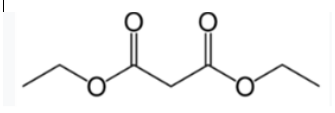
2. O/N 25/P22/Q4/d(i)

4(d)(i)	condensation	1
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3. O/N 25/P24/Q4/b(iii)

4(b)(iii)		reaction of V with propan-2-ol	3
	type of reaction	M1 condensation	
	functional group formed	M2 ester	
	molecular formula of organic product W	M3 C ₉ H ₁₈ O ₂	

4. M/J 25/P21/Q5/a,b,c(i),e(i,ii)

5(a)	M1 two ester links M2 rest of structure correct 	2
5(b)	(acid) hydrolysis	1
5(c)(i)	HOOCCH ₂ COOH + Na ₂ CO ₃ → NaOOCCH ₂ COONa + CO ₂ + H ₂ O	1
5(e)(i)	LiAlH ₄	1
5(e)(ii)	M1 correct unambiguous organic product M2 correct balanced equation HOOCCH₂COOH + 8[H] → HOCH₂CH₂CH₂OH + 2H₂O	2

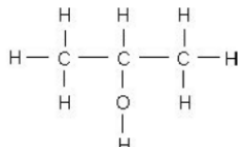
5. M/J 25/P22/Q4/a

4(a)(i)	LiAlH ₄	1
4(a)(ii)	$\text{CH}_3\text{CH}_2\text{CO}_2\text{H} + 4[\text{H}] \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{OH} + \text{H}_2\text{O}$	1

6. M/J 25/P23/Q5/b(ii,iii,iv)

5(b)(ii)		1
5(b)(iii)	reducing agent	1
5(b)(iv)	methanol / CH ₃ OH AND conc(entrated) (sulfuric) acid (catalyst)	1

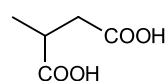
7. M/J 25/P24/Q4/b,c,d(ii)

4(b)	propyl ethanoate	1
4(c)(i)	condensation	1
4(c)(ii)	$\text{CH}_3\text{COOH} + \text{CH}_3\text{CH}_2\text{CH}_2\text{OH} \rightarrow \text{CH}_3\text{COOCH}_2\text{CH}_2\text{CH}_3 + \text{H}_2\text{O}$	1
4(c)(iii)	propan-1-ol	1
4(d)(ii)		2

8. M/J 24/P23/Q5/d(iii,iv)

5(d)(iii)	$\text{C}_8\text{H}_{12}\text{O}_4 + 8[\text{H}] \rightarrow \text{C}_8\text{H}_{16}\text{O}_2 + 2\text{H}_2\text{O}$	1
5(d)(iv)	LiAlH ₄	1

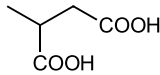
9. O/N 23/P21/Q2/c

2(c)	CH ₃ OH	1
		1
	one COOH group AND correct carbon framework (5C's) (each C has the correct valency)	
	both COOH groups	1

10. O/N 23/P22/Q3/c(iii),d(i)

3(c)(iii)	condensation	1
3(d)(i)	$\text{H}_3\text{PO}_4 + 3\text{CH}_3\text{OH} \rightarrow (\text{CH}_3\text{O})_3\text{PO} + 3\text{H}_2\text{O}$	1

11. O/N 23/P23/Q2/c

2(c)	CH ₃ OH	1
	 one COOH group AND correct carbon framework (5C's) (each C has the correct valency)	1
	both COOH groups	1

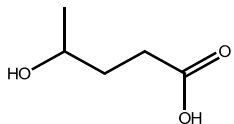
12. M/J 23/P21/Q5/c

5(c)(i)	suggest suitable reagent and conditions for step 1 heat AND any dilute named mineral acid e.g. dilute sulfuric acid / dilute hydrochloric acid	1
5(c)(ii)	 OR CH ₂ OHCH(OH)CH ₂ OH	1
5(c)(iii)	condensation	1
5(c)(iv)	methanol	1
5(c)(v)	methyl decanoate	1

13. M/J 23/P22/Q3/a(iv)

3(a)(iv)	$2 \text{ CH}_3\text{CH}_2\text{COOH} + \text{Ca} \rightarrow (\text{CH}_3\text{CH}_2\text{COO})_2\text{Ca} + \text{H}_2$ M1 H ₂ as product M2 balanced equation based on correct formula of Y	2
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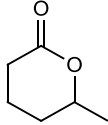
14. M/J 23/P22/Q5/d(i,ii,iii)

5(d)(i)	ester	1
5(d)(ii)	condensation	1
5(d)(iii)		1

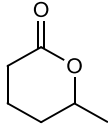
15. M/J 23/P23/Q3/c(i)

3(c)(i)	$\text{CH}_3\text{COOH} + \text{NaOH} \rightarrow \text{CH}_3\text{COONa} + \text{H}_2\text{O}$	1
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16. O/N 22/P21/Q5/a(iv)

5(a)(iv)		1
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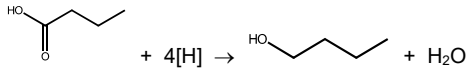
17. O/N 22/P23/Q5/a(iv)

5(a)(iv)		1
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18. M/J 22/P21/Q4/b(i)

4(b)(i)	M1 (add) group 1 carbonate / group 1 bicarbonate / Na_2CO_3 / NaHCO_3 etc. M2 effervescence / fizzing / bubbling	2
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19. M/J 22/P22/Q5/c(iii,iv)

5(c)(iii)	$\text{HOOC}(\text{CH}_2)_2\text{CH}_3 + 4[\text{H}] \rightarrow \text{HO}(\text{CH}_2)_3\text{CH}_3 + \text{H}_2\text{O}$ OR 	1
5(c)(iv)	LiAlH_4 OR lithium tetrahydridoaluminate (in ether)	1

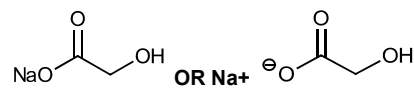
20. M/J 22/P22/Q6/a,c(iii)

6(a)	$\text{Br}_2(\text{aq})$	orange to colourless OR orange disappears	2
	$\text{Na}_2\text{CO}_3(\text{s})$	fizzing OR bubbles OR effervescence	
6(c)(iii)	OH O O OR O OH HO O		1

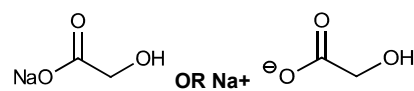
21. MAR 22/P22/Q4/b(i)

4(b)(i)	L = CH_3OH / methanol [1] conditions = acid(ic) / H^+ / H_2SO_4 AND (heat under) reflux [1]	2
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22. O/N 21/P21/Q4/c

4(c)(i)		1
4(c)(ii)	Not a strong (enough) reducing agent	1
4(c)(iii)	<i>Construct an equation</i> $(\text{CH}_2\text{OH})_2 + \text{SOCl}_2 \rightarrow (\text{CH}_2\text{Cl})_2 + \text{SO}_2 + \text{H}_2\text{O}$	1

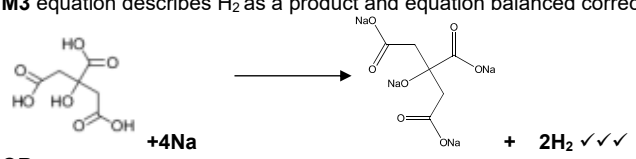
23. O/N 21/P23/Q4/c

4(c)(i)		1
4(c)(ii)	Not a strong (enough) reducing agent	1
4(c)(iii)	<i>Construct an equation</i> $(\text{CH}_2\text{OH})_2 + \text{SOCl}_2 \rightarrow (\text{CH}_2\text{Cl})_2 + \text{SO}_2 + \text{H}_2\text{O}$	1

24. M/J 21/P21/Q5/c(i)

5(c)(i)	CO_2H / carboxylic acid	1
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25. M/J 21/P23/Q5/a

5(a)	M1 all 3 COOH form COONa / CO ₂ Na / COO ⁻ Na ⁺	1
	M2 the OH group forms CONa / CO ⁻ Na ⁺	1
	M3 equation describes H ₂ as a product and equation balanced correctly  OR $\text{C}(\text{CH}_2\text{COOH})_2(\text{COOH})(\text{OH}) + 4\text{Na} \rightarrow \text{C}(\text{CH}_2\text{COONa})_2(\text{COONa})(\text{ONa}) + 2\text{H}_2 \checkmark\checkmark\checkmark$	1

26. M/J 20/P21/Q5/a(iii)

5(a)(iii)	M1 CO ₃ ²⁻ M2 propanoic acid – effervesce. (Propan-1-ol – no reaction)	2
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27. M/J 20/P21/Q6/f,g(ii)

6(f)(i)	$\text{CH}_3\text{CH}_2\text{CO}_2\text{H} + 4[\text{H}] \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{OH} + \text{H}_2\text{O}$	1
6(f)(ii)	propan-1-ol ALLOW propan-2-ol as error carried forward from 6f(i)	1
6(g)(ii)	Possible structure of W CH ₃ COOCH ₃ OR HCOOCH ₂ CH ₃	1

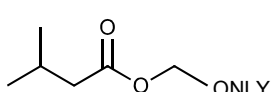
28. MAR 20/P22/Q3/a(v)

3(a)(v)	M1 hydrolysis M2 esterification / condensation	2
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29. O/N 19/P21/Q4/b(iv)

4(b)(iv)	 M1: / CH ₃ CH(OH)CH ₃ M2: H ₂ SO ₄ / sulfuric acid	2
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30. O/N 19/P22/Q4/d(iii)

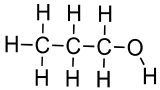
4(d)(iii)	M1: skeletal formula of Q  M2: one commercial use of Q (ethyl isovalerate / ethyl 3-methylbutyrate) solvents / perfumes / flavourings	2
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31. O/N 19/P23/Q4/b(iv)

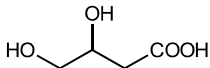
4(b)(iv)	 M1: / CH ₃ CH(OH)CH ₃ M2: H ₂ SO ₄ / sulfuric acid	2
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32. M/J 19/P22/Q5/a(i,ii,iii),d

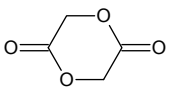
5(a)(i)	(n-)propyl ethanoate	1
5(a)(ii)	NaOH / sodium hydroxide	1

5a(iii)		1
5(d)	$2\text{HCO}_2\text{H} + \text{Na}_2\text{CO}_3 \rightarrow 2\text{HCO}_2\text{Na} + \text{CO}_2 + \text{H}_2\text{O}$	1

33. MAR 19/P22/Q4/c(ii,iv)

4(c)(ii)	M1 catalyst M2 ethanoic acid / $\text{CH}_3\text{CO}_2\text{H}$	2
4(c)(iv)	 M1 hydrolysed nitrile on straight-chain 4C backbone M2 3,4-diol	2

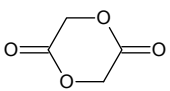
34. O/N 18/P21/Q4/d(ii)

4(d)(ii)	 M1 ANY ester group AND valid $\text{C}_4\text{H}_4\text{O}_4$ molecule M2 correct cyclic structure	2
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35. O/N 18/P22/Q4/e(i)

4(e)(i)	$\text{CH}_3\text{COCO}_2\text{H} + 6[\text{H}] \rightarrow \text{CH}_3\text{CHOHCH}_2\text{OH} + \text{H}_2\text{O}$ M1 correct organic product $\text{CH}_3\text{CHOHCH}_2\text{OH}$ [1] M2 [H] present as reactant with H_2O as product and balancing [1]	2
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36. O/N 18/P23/Q4/d(ii)

4(d)(ii)	 M1 ANY ester group AND valid $\text{C}_4\text{H}_4\text{O}_4$ molecule M2 correct cyclic structure	2
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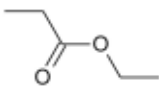
37. M/J 17/P22/Q4/b(ii)

4(b)(ii)	$2\text{C}_4\text{H}_8\text{O}_2 + \text{Na}_2\text{CO}_3 \rightarrow 2\text{C}_4\text{H}_7\text{O}_2\text{Na} + \text{H}_2\text{O} + \text{CO}_2$	1
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38. MAR 17/P22/Q3/d(i)

3(d)(i)	methyl pentanoate	1
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39. M/J 16/P21/Q5/b(i),d

(b) (i)	reduction	[1]	[1]
(d) (i)	(conc) H^+ / (conc) acid / (conc) H_2SO_4 / (conc) H_3PO_4	[1]	[1]
(ii)		[1]	[1]
(iii)	ethyl propanoate	[1]	[1]